TOPIC 08

Point Estimation

Using one number to estimate a parameter: estimators, bias, variance, maximum likelihood, and why we need intervals

WHAT YOU WILL LEARN

- Distinguish a parameter, an estimator and an estimate
- Judge estimators by bias, variance, consistency and efficiency
- Explain why the sample variance divides by n − 1
- Understand maximum likelihood estimation at an introductory level
- Explain why a point estimate alone is not enough

1. Parameter, estimator, estimate

Estimation is the first of the two main jobs of inference (the other is testing). We want to know a population **parameter** such as μ, σ² or p. We compute a single number from the sample and call it a **point estimate**.

Term	Meaning	Example
Parameter	Fixed, unknown population number	Mean income of all households, μ
Estimator	A *rule* (formula) applied to sample data; a random variable	$\bar{x} = (\Sigma x_i)/n$
Estimate	The *value* the rule gives for one particular sample	$\bar{x}$ = 31,250

The estimator is the recipe, the estimate is the cake. Different samples give different estimates from the same estimator.

Population parameter	Point estimator	Symbol
Mean	Sample mean	$\mu \to \bar{x}$
Variance	Sample variance	$\sigma^2 \to s^2$
Standard deviation	Sample standard deviation	$\sigma \to s$
Proportion	Sample proportion	$p \to \hat{p}$
Correlation	Sample correlation	$\rho \to r$

2. What makes an estimator good?

Bias

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

An estimator is **unbiased** if its average over all possible samples equals the true value (bias = 0). Unbiased does not mean any single estimate is right; it means the estimator is not systematically too high or too low.

Variance (precision)

Among unbiased estimators, prefer the one whose estimates are **less spread out**. The standard deviation of an estimator is its standard error (Topic 07).

Mean squared error combines both

$$\text{MSE}(\hat{\theta}) = E\left[(\hat{\theta} - \theta)^2\right] = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

Sometimes a slightly biased estimator with much lower variance has a smaller MSE than an unbiased one. This is the **bias-variance trade-off**, central in machine learning.

Consistency and efficiency

- **Consistent**: the estimate gets closer to the truth as n grows ($\bar{x} \to \mu$ by the Law of Large Numbers).
- **Efficient**: has the smallest variance among comparable unbiased estimators.
- **Sufficient**: uses all the information in the sample about the parameter ($\bar{x}$ does so for the mean of a normal population).

Property	Plain meaning	Dartboard picture
Low bias, low variance	Accurate and precise	Tight cluster on the bullseye
Low bias, high variance	Right on average, noisy	Scattered around the bullseye
High bias, low variance	Precisely wrong	Tight cluster away from the bullseye
High bias, high variance	Worst case	Scattered and off-centre

3. Important estimators and their properties

Estimator	Unbiased?	Remarks
Sample mean $\bar{x}$ for μ	Yes	Consistent; minimum-variance for normal data; SE = $\sigma/\sqrt{n}$
Sample proportion $\hat{p}$ for p	Yes	SE = $\sqrt{p(1-p)/n}$
Sample variance s^2 (divide by $n-1$) for σ^2	Yes	The standard choice
Variance with divide-by-n for σ^2	**No** (underestimates)	Bias factor $(n-1)/n$; vanishes as n grows
Sample standard deviation s for σ	Slightly biased (low)	Tiny bias; accepted because $\sqrt{}$ of an unbiased variance
Sample median for μ (normal data)	Yes (symmetric)	Less efficient: its variance is about $\pi/2 \approx 1.57$ times that of $\bar{x}$

4. Why divide by n − 1?

Dividing the squared deviations by n gives a variance estimate that is **too small on average**. The reason is that deviations are measured from $\bar{x}$, which is itself computed from the same data and is the point that makes the squared deviations as small as possible. Measuring from the *true* μ would give larger values. Dividing by **n − 1** (the **degrees of freedom**) exactly corrects the shortfall. Once $\bar{x}$ is known, only $n-1$ of the deviations are free to vary, because they must sum to zero.

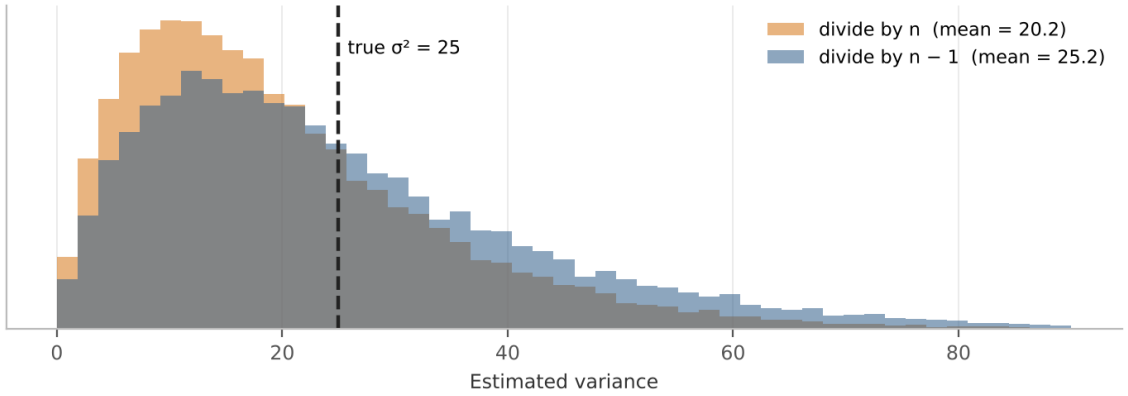


Figure 1. 20,000 samples of size 5 from a population with $\sigma^2 = 25$. Dividing by n underestimates on average (mean 20.2); dividing by $n - 1$ is unbiased (mean 25.2).

Degrees of freedom in one line. Take four values with mean $\bar{x}$. Their deviations from $\bar{x}$ must add to 0. If three deviations are -3, -1 and 1, the fourth is forced to be 3: only $n - 1 = 3$ deviations are free to vary, and that is the number we divide by.

The simulation agrees with theory: the n-divisor estimator averages about $(n - 1)/n \times 25 = 20$, the $n - 1$ divisor averages about 25.

5. Mean versus median as an estimator

For normal data both estimate the centre without bias, but the mean is more efficient. In a simulation of 20,000 samples of size 25 the variance of the sample median was 1.53 times that of the sample mean (theory: $\pi/2 \approx 1.57$ for large n). The mean gets the same precision from fewer observations. For data with outliers or heavy tails the median can win, a reminder that 'best' depends on the population.

6. How are estimators found? (introduction)

Method of moments

Match sample moments to population moments. The population mean μ is estimated by the sample mean; the second moment gives a variance estimator. It is simple but not always the most efficient. Example: the first moment of the population is $E(X) = \mu$, and the first sample moment is $\bar{x}$, so setting them equal gives the estimator $\hat{\mu} = \bar{x}$.

Maximum likelihood estimation (MLE)

Choose the parameter value that makes the **observed data most probable**. The **likelihood** $L(\theta)$ is the probability (density) of the observed data viewed as a function of the unknown parameter θ.

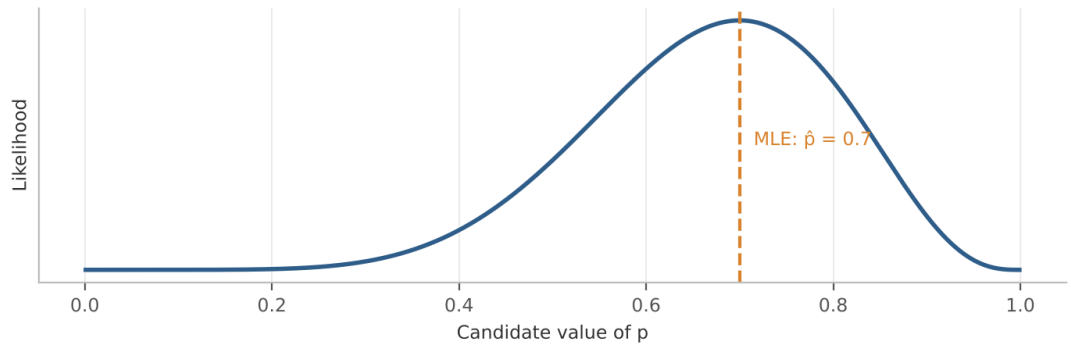


Figure 2. Likelihood of p after observing 7 heads in 10 flips. It peaks at p = 0.7, the sample proportion.

WORKED EXAMPLE **Coin flips**

We flip a coin 10 times and see 7 heads. The likelihood of p is $L(p) = C(10,7)\, p^7 (1-p)^3$. $L(0.5) = 0.1172$, but $L(0.7) = \mathbf{0.2668}$, higher. Calculus (or Figure 2) shows the maximum is at $p = 7/10$, so the MLE equals the sample proportion.

For a normal population the MLEs are $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \Sigma(x_i - \bar{x})^2/n$ (divide by n, slightly biased). MLEs are consistent and, for large n, nearly efficient and approximately normal, which is why they dominate modern statistics and machine learning.

7. The limit of point estimates

A point estimate gives no hint of **how far off it might be**. Suppose you visit 5% of the restaurants in London and find that the average meal costs £22.50. You may be close, but the true average is almost certainly not exactly 22.50; something between £20 and £25 is more believable. Reporting **[20, 25]** says far more about reality than the single number 22.50 does.

KEY IDEA **From a point to a range**

A point estimate is the best single guess but carries no measure of uncertainty. A **confidence interval** adds that uncertainty using the standard error: point estimate ± margin of error. That is Topic 09.

Point estimate versus interval

	Point estimate	Interval estimate
What it gives	One number	A range of plausible values
Chance it is exactly right	Essentially 0 for continuous data	Controlled (e.g. 95%)
Uncertainty shown?	No	Yes, through its width
Example	$\bar{x} = 22.50$	[20, 25] with 95% confidence

8. In Python

```
import numpy as np
from scipy import stats

x = np.array([12.1, 14.3, 11.8, 15.2, 13.7, 12.9, 14.8, 13.1])
print(x.mean())                  # estimate of mu
print(x.var(ddof=1))             # unbiased estimate of sigma^2 (ddof=0 divides by n)

# MLE of a coin's p from 7 heads in 10 flips = sample proportion
print(7 / 10)

# MLE for a normal fit
mu_hat, sigma_hat = stats.norm.fit(x)    # sigma_hat uses divisor n
print(mu_hat, sigma_hat)
```

SUMMARY

- Parameter (fixed, unknown) ≠ estimator (random rule) ≠ estimate (one realised value).
- Good estimators have low bias and low variance; MSE = variance + bias².
- $\bar{x}$, $\hat{p}$ and s^2 (with $n - 1$) are unbiased; dividing by n underestimates σ^2.
- MLE picks the parameter value that makes the data most likely.
- A point estimate hides uncertainty, so we add an interval (Topic 09).

Practice questions

Try each question before reading the answer underneath it.

Q1. What is the difference between an estimator and an estimate?

Answer. An estimator is the formula (random, varies by sample); an estimate is the number it produces for one sample.

Q2. Data: 4, 6, 8, 10. Find $\bar{x}$ and the unbiased s^2.

Answer. $\bar{x} = 7$; deviations $-3, -1, 1, 3$, squares sum to 20; $s^2 = 20/3 = 6.667$ (divide by n would give 5).

Q3. Why is dividing by n biased for variance?

Answer. Deviations are measured from $\bar{x}$, which is fitted to the same data, so they are smaller than deviations from μ; dividing by $n - 1$ corrects this.

Q4. Estimator A: bias 0, variance 9. Estimator B: bias 1, variance 4. Which has the smaller MSE?

Answer. A: 9. B: $4 + 1^2 = 5$. B has the smaller MSE.

Q5. Out of 200 customers, 38 are repeat buyers. Give the point estimate and say what it cannot tell you.

Answer. $\hat{p} = 38/200 = 0.19$. It does not show how far the true proportion might be from 0.19.

Q6. You observe 3 heads in 12 flips. What is the MLE of p?

Answer. $\hat{p} = 3/12 = 0.25$.